StepUp Final Project Report

Fitness and Physical Activity Tracking Platform

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Note on status words. Implemented, **Partially implemented** and **Planned** describe each feature. Italic text in square brackets, [*like this*], is a placeholder for evidence the team must supply; no value has been invented. Status is read from the project timeline ("developed" or "integrated" = Implemented; "worked on" = Partially implemented) and must be checked against the final build.

1. Title Page, Declaration and Acknowledgement

Declaration. This report on StepUp is an original record of our project work and has not been submitted for any other degree or diploma. All sources used are listed in the References chapter.

Acknowledgement. We thank our project guide and the faculty of the department for their guidance, the peers who tested the application, and our families for their support.

2. Abstract

Fitness applications that rank users and reward step counts assume that the counted steps were actually walked. A shaken phone, vehicle vibration, or step records written by another application can inflate a total and make leaderboards and rewards unfair.

StepUp is an Android fitness platform for students. It offers daily step tracking, a dashboard, fitness goals, challenges, step battles, a leaderboard, a sport buddy, activity sessions, and a wallet with reward points. Its distinguishing feature is an activity-integrity evaluation: Android Health Connect records are treated as one input and cross-checked against on-device motion evidence, not as proof of walking.

The application uses Flutter, native Android sensors, Health Connect, and a REST backend with PostgreSQL. This report documents the architecture, the layered verification design, the modules, the test plan and the challenges met. Modules are marked Implemented, Partially implemented or Planned. Measured results are not reported; they are marked placeholders. Server-side risk checks and cloud deployment are future work.

3. Introduction and Problem Statement

3.1 Background

Walking is the simplest form of exercise, and smartphones make it measurable: Android phones carry motion sensors and a hardware step counter, and Health Connect lets applications read and write health records such as steps in one shared store. Motivation is the harder problem. Comparison with friends, shared challenges and rewards give students a reason to return daily, and StepUp is built around this idea.

3.2 Problem Statement

When steps decide rankings and rewards, the accuracy of the count becomes a fairness issue. Three weaknesses are common:

- **Motion that is not walking:** shaking a phone or riding in a vehicle can register steps on a motion-based counter.
- **Records from other sources:** Health Connect holds records written by several applications, and a record can be entered manually or written by a step-generator app [5]. A Health Connect total does not prove that the user walked.
- **Timing and background limits:** the step counter reports the time of the last step and may deliver it up to about ten seconds late [6], and Android restricts sensor events for background apps [8].

Health Connect alone therefore cannot be treated as a guarantee of genuine or fraud-free steps. On a phone its step data comes from the same motion hardware, so shaken-phone steps can still appear in it.

3.3 Proposed Solution

StepUp combines a social fitness experience with an activity-integrity evaluation. It uses Health Connect and on-device sensors together, separates steps that are only displayed from steps eligible for rewards, and limits credit to what motion evidence supports.

4. Project Objectives and Scope

4.1 Objectives and Outcome

Objectives are re-assessed against the final state of the work. “Met” means the module is recorded as implemented.

ID	Objective	Outcome	Remarks
O1	Track daily steps with distance, calories and goal progress on a dashboard	Met	Weeks 6–7
O2	Count walking steps from phone sensors in real time	Met	Week 7
O3	Detect and reject fake steps (shaking, vehicle travel)	Partially met	Design complete; layer status [*confirm*]
O4	Use Health Connect as a data source and cross-check	Partially met	Week 9
O5	Student leaderboard and social competition	Met	Weeks 10–11
O6	Challenges, milestones, badges, reward points	Partially met	Week 12
O7	Wallet, activity sessions, notifications	Met	Week 13
O8	Integrate all modules and test with users	Planned	Week 14 [*confirm*]

Table 1: Project objectives and outcome

4.2 Scope

In scope: an Android app for students; step tracking and integrity evaluation; the modules in Chapter 9; a REST backend and database. **Out of scope:** iOS, wearable import, Play Store release, a production risk engine, and cloud deployment (future work, Chapter 13). The report does not claim that every form of cheating is prevented.

4.3 Changes from the Original Proposal

The project website records two initial tracks: activity-integrity detection and sport-group matching. The final system keeps integrity evaluation at its core and treats buddy and group features as social modules built on verified steps. [*Add any other scope change and the reason.*]

5. Literature Review and Existing Systems

5.1 Smartphone Step Counting

Brajdic and Harle compared walk-detection and step-counting algorithms on unconstrained smartphones, carried in hand, pocket or bag [1]. Kang et al. proposed a gyroscope-based method that first decides whether the user is walking and then estimates step frequency [2]. Khedr and El-Sheimy fused inertial and magnetometer data in an adaptive step counter [3]. These works address accurate counting, not a user who tries to fake steps, which is the gap StepUp targets. Published accuracy figures depend on their data and phone placement and are not properties of StepUp.

5.2 Platform Capabilities and Existing Applications

Feature	What it provides	Implication for StepUp
Hardware step counter	Cumulative count; timestamp is the last step, delivery may lag about 10 s [6]	Bucket by sensor timestamp; settle late events
Health Connect	Shared store; each record has source metadata and a recording method (automatic, active, manual) [5]	Use as input and cross-check, never as proof
Foreground service	Continuous sensing under a visible notification [7]	Needed for background capture
Play Integrity	Device and app integrity signals [9]	Raises attacker cost only

Table 2: Platform features and their implication

General-purpose trackers display counts from the phone, a wearable or Health Connect and generally accept the incoming count, because their aim is personal tracking. Platforms built on ranking and rewards depend far more on the validity of that count. [*Add two or three named applications the team reviewed, with a short factual comparison.*]

6. System Requirements and Technology Stack

6.1 Requirements

- **Functional:** registration, login and profile; dashboard with steps, distance, calories and goals; real-time step detection; integrity evaluation; Health Connect permissions, retrieval and sync; leaderboard, challenges and step battles; rewards, wallet, sessions and notifications.
- **Integrity:** steps that cannot be verified must not earn rewards.
- **Reliability:** capture should continue in the background, subject to Android limits.
- **Privacy:** health data is read only with user permission; only data needed for ranking and rewards is sent

to the server. [*State the actual data-handling policy.*]

- **Performance and battery:** to be measured; no targets are claimed.

6.2 Technology Stack

Layer	Technology	Purpose	Status
Mobile UI	Flutter (Dart)	Android application interface	**Implemented**
Sensors	Accelerometer, gyroscope, step counter	Motion data and cumulative steps	**Implemented**
Health data	Android Health Connect	Step records and cross-check	**Partially implemented**
Native layer	Kotlin foreground service	Background sensing	[*confirm*]
Backend	REST API [*Node.js/Express: confirm*]	Auth, challenges, rankings, rewards	[*confirm*]
Database	PostgreSQL [*with Prisma: confirm*]	Users, verified steps, wallet	[*confirm*]
Hosting	AWS	Backend hosting	**Planned**

Table 3: Technology stack

[*List the actual test devices (model, Android version), IDE, Flutter and SDK versions. Health Connect must be available on the test device.*]

7. System Architecture and Design

7.1 Overview

The system has three parts: data sources on the phone, the application with its step-integrity pipeline, and a backend with a database (Figure 1). Only verified step evidence reaches the backend, which remains the authority for rankings and rewards.

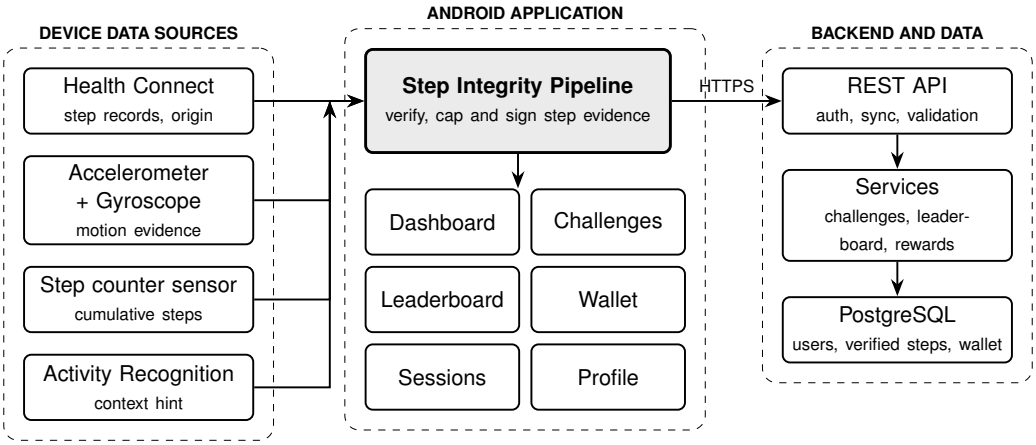


Figure 1: High-level system architecture

7.2 Layered Step-Verification Pipeline

The initial proposal described six layers. After review it was refined into the seven layers of Figure 2. The principal change is that Health Connect no longer decides the reward count: motion evidence from the device limits what can be credited.

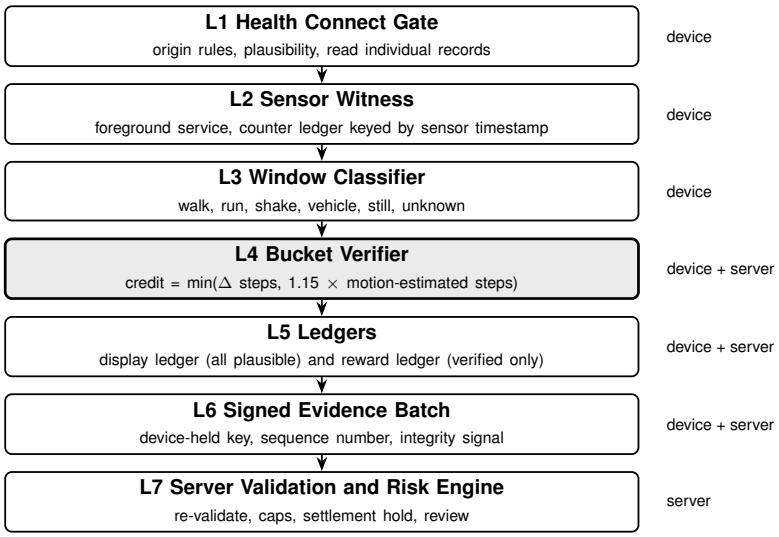


Figure 2: Layered step-verification pipeline

Layer	Responsibility	Status
L1 Health Connect Gate	Accept only allowed sources; reject implausible values; read individual records	Designed; code drafted [*confirm*]
L2 Sensor Witness	Record counter and motion data in a foreground service	Designed [*confirm*]
L3 Window Classifier	Label short windows (walk, run, shake, vehicle, still)	Designed [*confirm*]
L4 Bucket Verifier	Credit steps per bucket, capped by motion evidence	Designed; code and tests drafted, not run [*confirm*]
L5 Ledgers	Display ledger and reward ledger	Designed [*confirm*]
L6 Signed batch	Sign evidence with a device-held key	**Planned**
L7 Server validation	Re-validate, apply caps and hold, flag anomalies	**Planned**

Table 4: Layer responsibilities and status

8. Methodology and Implementation

8.1 Development Approach

The project followed a 15-week incremental plan with one milestone per week. Weeks 1 to 13 are recorded as completed; Weeks 14 and 15 are planned at the time of writing. [*Update if completed before submission.*]

Phase	Weeks	Outcome
Foundation	1–3	Problem definition, research, requirements, architecture and technology choice
App build	4–6	Interface, navigation, authentication, profile, dashboard
Integrity	7–9	Sensor integration, layered anti-fraud design, Health Connect sync
Social	10–13	Leaderboard, buddy and individual challenges, rewards, wallet, sessions, notifications
Delivery	14–15	Integration, testing, optimisation, documentation, demonstration (planned)

Table 5: Project phases

8.2 Step-Crediting Method

Figure 3 shows how a ten-second bucket is handled. Steps are never rewarded merely because a counter increased. They are rewarded only when the record source is acceptable and the motion evidence looks like

walking, and credit is capped at 1.15 times the gait steps estimated from motion sensors. This limits what a mixed walk-and-shake window can gain.

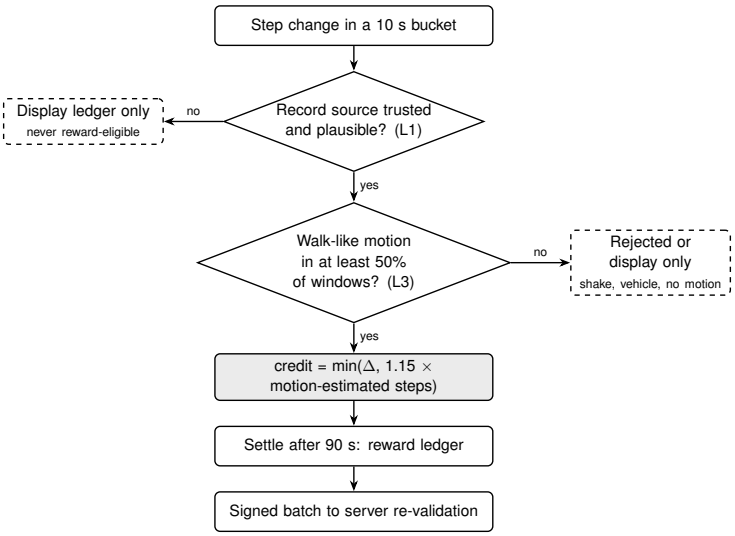


Figure 3: Step-crediting decision flow

Initial design values: bucket length 10 s, settlement delay 90 s, walk-like share of windows $\geq$ 50%, credit cap 1.15 $\times$ motion steps. These are starting points to be tuned with the team's own data. [*Record the final values used in the build.*]

8.3 Implementation Notes

- **Application:** Flutter screens for the dashboard, challenges, leaderboard, wallet, sessions and profile.
- **Health Connect:** the app requests step-read permission and retrieves step records, which are compared with the app's own count and never used alone to grant rewards.
- **Sensors:** accelerometer and gyroscope data provide candidate steps and the motion evidence for the verifier.
- **Backend:** a REST API handles accounts, challenges, rankings and rewards. [*Add endpoint list or API document reference.*]

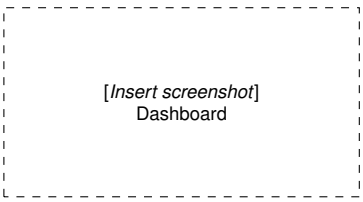
9. Module-wise Description

Module	Function	Week	Status
Authentication and profile	Registration, login, student profile	5	**Implemented**
Dashboard and tracking	Steps, distance, calories, goals, summary	6	**Implemented**
Sensor integration	Accelerometer and gyroscope; real-time candidate steps	7	**Implemented**
Activity integrity evaluation	Layered validation against shaking and vehicle travel	8	**Partially implemented**
Health Connect integration	Permissions, record retrieval, sync	9	**Partially implemented**
Leaderboard	Rankings by verified steps	10	**Implemented**
Buddy challenges	Challenges with friends, shared progress	11	**Implemented**
Individual challenges and rewards	Daily and weekly challenges, milestones, badges, points	12	**Partially implemented**
Wallet, sessions, notifications	Points wallet, session history, alerts	13	**Implemented**
Step Battles	Competitive step comparison between users	–	[*confirm status*]
Sport Buddy	Finding and pairing with a fitness partner	–	[*confirm status*]
Risk engine, signed evidence	Server re-validation, caps, anomaly review	–	**Planned**

Table 6: Module summary (status follows the project timeline; verify against the final build)

Activity integrity evaluation. A plausible step is shown on the dashboard, but it enters the reward ledger only when verified. Steps recovered after a gap in sensing are displayed but never rewarded. The module does not claim to catch every cheat: a motorised shaking rig or a modified phone may still earn credit within the caps (Chapter 13).

Social and reward modules. The leaderboard ranks users by verified steps; buddy challenges, individual challenges and milestones give shared and personal targets; points accumulate in the wallet; sessions record bounded periods of exercise; notifications report challenge and goal events.





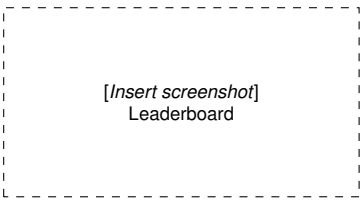


Figure 4: Application screenshots (replace placeholders with actual screens)

10. Testing and Validation

Testing covers functional tests per module, integrity tests with deliberate fake-step scenarios, and user acceptance testing with students. Reference unit tests for the bucket-verification rule were drafted during design, but their results have not been recorded. **No test results are reported here;** result cells are placeholders for the team's own logs.

ID	Test scenario	Expected behaviour	Result
T1	Normal walking, phone in hand and pocket	Steps counted and verified	[*pass/fail, count*]
T2	Running	Counted; classified as run	[ ]
T3	Phone shaken while stationary	Not credited to reward ledger	[ ]
T4	15 s walk then 15 s shake	Credit limited to genuine walking	[ ]
T5	Travelling in a vehicle	Not credited	[ ]
T6	Manual or third-party Health Connect record	Never reward-eligible	[ ]
T7	Health Connect permission denied	Graceful message and fallback	[ ]
T8	App in background or removed from recents	Capture continues, or gap is display-only	[ ]
T9	Challenge completion and reward credit	Points added once	[ ]
T10	Leaderboard after new steps	Ranking reflects verified steps	[ ]
T11	Login, registration, profile edit	Data saved and retrieved	[ ]
T12	Network or backend unavailable	Local data kept; sync resumes	[ ]

Table 7: Test plan and result record

User acceptance testing. [*State number of testers, how chosen, tasks given, devices and feedback. Do not report figures that were not measured.*] **Performance and battery.** [*Record device model, test duration and measured response time, battery drain and crashes.*]

11. Results and Analysis

11.1 Development Outcome

Weeks 1 to 13 produced the interface, accounts, dashboard, sensor integration, the integrity design, Health Connect integration work, leaderboard, challenge modules, wallet, sessions and notifications. Of the twelve rows in the module table, six are Implemented, three Partially implemented, two awaiting confirmation and one Planned, according to the timeline wording.

11.2 Integrity Evaluation

These measures should be reported once tests T3 to T6 are run. They are empty because no measurements were supplied.

Measure	Definition	Value
Genuine-step retention	Credited steps divided by hand-counted steps in walking tests	[*measured*]
Shake rejection	Steps from stationary shaking not credited	[*measured*]
Vehicle rejection	Steps in a vehicle not credited	[*measured*]
Mixed-window credit ratio	Credited divided by genuine steps in walk-plus-shake tests	[*measured*]

Table 8: Integrity evaluation (to be completed)

11.3 Analysis

If implemented as described, the design gives specific guarantees: records from other apps and manual entries never earn rewards; a counter that rises without matching motion earns nothing; and credit in mixed windows is bounded by the motion estimate. These are properties of the design, not measured outcomes.

The design also has limits. Health Connect cannot detect shaking by itself, so on-device motion evidence is essential. A human-like mechanical shaker, a phone carried by an animal, or a rooted device that tampers

with sensors could still earn credit, which is why daily caps, a settlement hold and server review are planned. Until they exist, the checks should be described as reducing fake steps, not eliminating them.

12. Challenges and Solutions

Challenge	Approach adopted	Status
Health Connect cannot prove walking	Cross-check only; cap credit by motion evidence; allow trusted sources	Designed; partly implemented
Late step-counter events	Key steps by sensor timestamp; settle buckets after a delay	Designed
Shaking and vehicle motion	Classify short windows; cap credit by estimated gait steps	Designed; partly implemented
Background execution limits	Foreground service; gaps are display-only	[*confirm*]
Permissions and availability	Request at use; clear messages if unavailable	**Partially implemented**
Integrating many modules	Weekly increments; final integration week	**Planned** (Week 14)

Table 9: Challenges and approach adopted

[*Add the hardest technical problem the team actually faced and how it was solved.*]

13. Conclusion and Future Scope

StepUp pairs a student-focused social fitness experience with a layered approach to verifying steps. Accounts, dashboard, sensor-based detection, leaderboard, challenges, wallet and sessions are recorded as built; integrity evaluation and Health Connect integration are partially implemented. The key decision is to trust no single source: Health Connect, the step counter and motion sensors are used together, and only verified steps earn rewards.

Limitations. Test, accuracy and performance figures are not yet measured; server validation, signed evidence and anomaly review are not in place; the checks reduce fake steps but cannot stop a determined attacker with specialised equipment.

Future scope.

- Complete the on-device layers, run the Chapter 10 tests and calibrate the design values.
- Add signed evidence batches and a server risk engine with caps, settlement hold and review.
- **Cloud deployment:** hosting the backend on AWS is planned future work; no deployment evidence is included.
- Add wearable import with lower weight after separate validation, and run a controlled student pilot.

14. References

[1] A. Brajdic and R. Harle, "Walk detection and step counting on unconstrained smartphones," in *Proc. ACM UbiComp '13*, 2013, doi: 10.1145/2493432.2493449.

[2] X. Kang, B. Huang and G. Qi, "A Novel Walking Detection and Step Counting Algorithm Using Unconstrained Smartphones," *Sensors*, vol. 18, no. 1, art. 297, 2018.

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[4] Android Developers, "Track steps," Health Connect. `https://developer.android.com/health-and-fitness/health-connect/features/steps`

[5] Android Developers, “Write data,” Health Connect. https://developer.android.com/health-and-fitness/health-connect/write-data

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[*Add any additional sources the team used.*]